Function	#	Description	Sample Input Data	Expected Output	Actual Output	P/F
compareStrings	1	Identical uppercase strings	string1 = "AE104", string2 = "AE104"	Return 0	Return 0	**P**
	2	Identical lowercase strings	string1 = "ae156", string2 = "ae156"	Return 0	Return 0	**P**
	3	Identical mixed-case strings	string1 = "Spec1", string2 = "Spec1"	Return 0	Return 0	**P**
	4	Identical strings with different capitalization	string1 = "spec1", string2 = "SPEC1"	Return 0	Return 0	**P**
	5	string1 comes before string2 alphabetically	string1 = "AE104", string2 = "SPEC1"	Return int < 0	Return -18	**P**
	6	string1 comes after string2 alphabetically	string1 = "spec2", string2 = "ae152"	Return int > 0	Return 18	**P**
isValidTrip	1	Valid regular trip in uppercase	tripNo = "AE105"	Return 4	Return 4	**P**
	2	Valid regular trip in lowercase	tripNo = "ae152"	Return 12	Return 12	**P**
	3	Valid special trip in uppercase	tripNo = "SPEC1"	Return 9	Return 9	**P**
	4	Valid special trip in lowercase	tripNo = "spec2"	Return 21	Return 21	**P**
	5	Invalid regular trip	tripNo = "spec3"	Return -1	Return -1	**P**
	6	Invalid special trip	tripNo = "ae520"	Return -1	Return -1	**P**
isValidID	1	Valid and unique 8-digit ID	idNo = 12345678	Return 1	Return 1	**P**
	2	Valid and unique non-8-digit ID	idNo = 1234	Return 1	Return 1	**P**
	3	Zero ID	idNo = 0	Return 0	Return 0	**P**
	4	Negative ID	idNo = -123	Return 0	Return 0	**P**
	5	Valid but duplicate ID	idNo = 12345678	Return 2	Return 2	**P**
isFullTrip	1	Empty trip	AE101 empty, priorityNo = 3	Return 0	Return 0	**P**
	2	Last seat of trip vacant	AE101 last seat empty, priorityNo = 2	Return 0	Return 0	**P**
	3	Full regular trip and new passenger has lower priority	AE151 last passenger priority = 2, priorityNo = 3	Return 2	Return 2	**P**
	4	Full regular trip and new passenger has higher priority	AE151 last passenger priority = 2, priorityNo = 1	Return 1	Return 1	**P**
	5	Full special trip and new passenger has lower priority	SPEC2 last passenger priority = 3, priorityNo = 5	Return 2	Return 2	**P**
	6	Full special trip and new passenger has higher priority	SPEC2 last passenger priority = 3, priorityNo = 1	Return 1	Return 1	**P**
isSpecDeployed	1	Only SPEC1 has at least one passenger	SPEC1 has one passenger, SPEC2 empty	Return 1	Return 1	**P**
	2	Only SPEC2 has at least one passenger	SPEC1 empty, SPEC2 has one passenger	Return 2	Return 2	**P**
	3	SPEC1 and SPEC2 have no passengers	SPEC1 empty, SPEC2 empty	Return 0	Return 0	**P**
	4	SPEC1 and SPEC2 both have passengers	SPEC1 has one passenger, SPEC2 has one passenger	Return 3	Return 3	**P**
	5	All trips have no passengers	AE101 empty, SPEC1 empty, SPEC2 empty	Return 0	Return 0	**P**
convertDropOff	1	Same drop-off set	dropOffOld = 4, dropOffSetOld = 2, dropOffSetNew = 2	Return 4	Return 4	**P**
	2	Convert existing drop-off from drop-off set 0 to 1	dropOffOld = 3, dropOffSetOld = 0, dropOffSetNew = 1	Return 2	Return 2	**P**
	3	Convert existing drop-off from drop-off set 1 to 0	dropOffOld = 2, dropOffSetOld = 1, dropOffSetNew = 0	Return 3	Return 3	**P**
	4	Convert existing drop-off from drop-off set 2 to 3	dropOffOld = 2, dropOffSetOld = 2, dropOffSetNew = 3	Return 2	Return 2	**P**
	5	Convert existing drop-off from drop-off set 3 to 2	dropOffOld = 4, dropOffSetOld = 3, dropOffSetNew = 2	Return 4	Return 4	**P**
	6	Convert non-existent drop-off from drop-off set 0 to 1	dropOffOld = 2, dropOffSetOld = 0, dropOffSetNew = 1	Return 2	Return 2	**P**
	7	Convert non-existent drop-off from drop-off set 1 to 0	dropOffOld = 1, dropOffSetOld = 1, dropOffSetNew = 0	Return 3	Return 3	**P**
	8	Convert non-existent drop-off from drop-off set 2 to 3	dropOffOld = 1, dropOffSetOld = 2, dropOffSetNew = 3	Return 4	Return 4	**P**
	9	Convert non-existent drop-off from drop-off set 3 to 2	dropOffOld = 1, dropOffSetOld = 3, dropOffSetNew = 2	Return 4	Return 4	**P**
insertPassenger	1	Empty trip	priorityNo = {99, ..., 99}, temp.priorityNo = 2	priorityNo = {2, 99, ..., 99}	priorityNo = {2, 99, ..., 99}	**P**
	2	Trip is not full and has one higher priority passenger	priorityNo = {1, 99, ..., 99}, temp.priorityNo = 2	priorityNo = {1, 2, 99, ..., 99}	priorityNo = {1, 2, 99, ..., 99}	**P**
	3	Trip is not full and has one lower priority passenger	priorityNo = {3, 99, ..., 99}, temp.priorityNo = 2	priorityNo = {2, 3, 99, ..., 99}	priorityNo = {2, 3, 99, ..., 99}	**P**
	4	Trip is not full and has mixed passengers	priorityNo = {1, 2, 2, 5, 99, ..., 99}, temp.priorityNo = 2	priorityNo = {1, 2, 2, 2, 5, 99, ..., 99}	priorityNo = {1, 2, 2, 2, 5, 99, ..., 99}	**P**
	5	Trip is full and new passenger has lower priority	priorityNo = {1, ..., 1}, temp.priorityNo = 2	priorityNo = {1, ..., 1}	priorityNo = {1, ..., 1}	**P**
	6	Trip is full and new passenger has higher priority	priorityNo = {3, 3, 4, ..., 5, 6}, temp.priorityNo = 2	priorityNo = {2, 3, 3, 4, ..., 5}	priorityNo = {2, 3, 3, 4, ..., 5}	**P**
	7	Trip is full and has mixed passengers	priorityNo = {1, 2, 2, 3, 5, ..., 6, 6}, temp.priorityNo = 2	priorityNo = {1, 2, 2, 2, 3, 5, ..., 6}	priorityNo = {1, 2, 2, 2, 3, 5, ..., 6}	**P**
movePassenger	1	Move last passenger to next trip (empty)	Current trip = {1, ..., 1}, Next trip = {99, ..., 99}	Next trip = {1, 99, ..., 99}	Next trip = {1, 99, ..., 99}	**P**
	2	Move last passenger to next trip (not empty)	Current trip = {1, ..., 1}, Next trip = {1, 99, ..., 99}	Next trip = {1, 1, 99, ..., 99}	Next trip = {1, 1, 99, ..., 99}	**P**

Function	#	Description	Sample Input Data	Expected Output	Actual Output	P/F
	3	Move last passenger to next trip (full)	Current trip = {1, ..., 1}, Next trip = {1, ..., 1}	Next trip = {1, ..., 1}	Next trip = {1, ..., 1}	**P**
	4	Move last passenger to special trip (empty)	Current trip = {1, ..., 1}, Special trip = {99, ..., 99}	Special trip = {1, 99, ..., 99}	Special trip = {1, 99, ..., 99}	**P**
	5	Move last passenger to special trip (not empty)	Current trip = {1, ..., 1}, Special trip = {1, 99, ..., 99}	Special trip = {1, 1, 99, ..., 99}	Special trip = {1, 1, 99, ..., 99}	**P**
	6	Move last passenger to special trip (full)	Current trip = {1, ..., 1}, Special trip = {1, ..., 1}	Special trip = {1, ..., 1}	Special trip = {1, ..., 1}	**P**
	7	Move last passenger from special trip	Special trip = {1, ..., 1}, Next trip (diff. route) = {99, ..., 99}	Next trip (diff. route) = {99, ..., 99}	Next trip (diff. route) = {99, ..., 99}	**P**
getDate	1	Valid date	mm = 3, dd = 31, yyyy = 2025	Return {31, 03, 2025}	Return {31, 03, 2025}	**P**
	2	Invalid month	mm = 14	printError()	printError()	**P**
	3	Invalid day	dd = 34	printError()	printError()	**P**
	4	Invalid year	yyyy = 2026	printError()	printError()	**P**